Statistical Learning – Classification

STAT 841 / CM 763 – Fall 2026

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4:00–5:20 p.m.   MC 4042

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Course Outline Notice

- Course content, topic order, assessment details, dates, deadlines, and other arrangements are tentative and may change.
- Any changes will be announced through the official course platforms.

Statistics and Machine Learning

- **Shared goal:** Learn from data and generalize beyond the observed sample.
- **Statistical emphasis:** Models, assumptions, estimation, uncertainty, and inference.
- **Machine-learning emphasis:** Prediction, scalable computation, and learned representations.
- **Statistical learning:** Combines these perspectives rather than separating them.

Core Learning Problems

- **Classification:** Predict a discrete label Y.
- **Regression:** Predict a continuous response Y.
- **Clustering:** Discover groups without observed labels.
- **Representation learning:** Construct informative lower-dimensional features from X.

Classification

Consider data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $\mathbf{x} \in \mathbb{R}^d$ and y_i **takes values in some finite set**.

Find a function f,such that when we observe a new $\mathbf{x}$ we predict y to be $f(\mathbf{x})$.

Regression

Consider data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $\mathbf{x} \in \mathbb{R}^d$ and y_i **takes values in** $\mathbb{R}$.

Find a function f, such that when we observe a new $\mathbf{x}$ we predict y to be $f(\mathbf{x})$.

Clustering

Consider data $\{\mathbf{x}_i\}_{i=1}^{n}$ where $\mathbf{x} \in \mathbb{R}^d$.

Find a function f, when we observe a new $\mathbf{x}$ we predict y to be $f(\mathbf{x})$, such that for similar $\mathbf{x}$, y is the same.

Learning Settings

- **Supervised learning:** Learn from input–output pairs; examples include classification and regression.
- **Unsupervised learning:** Find structure from inputs alone; examples include clustering and dimensionality reduction.
- **Semi-supervised learning:** Combine a small labelled set with a larger unlabelled set.

Classification in Healthcare

- **Clinical risk prediction:** Estimate whether a patient will develop a condition or experience an adverse event.
- **Medical-image triage:** Classify scans by whether they require urgent review.
- **Treatment-response prediction:** Identify which patients are likely to benefit from an intervention.

Classification in Finance

- **Credit risk:** Predict default or delinquency categories.
- **Fraud detection:** Classify transactions or accounts as suspicious or legitimate.
- **Alert triage:** Prioritize compliance and anti-money-laundering investigations.

Classification in Technology

- **Spam and abuse detection:** Classify messages, accounts, or content.
- **Language understanding:** Predict sentiment, topic, intent, or toxicity.
- **Visual recognition:** Assign labels to images, objects, or video segments.

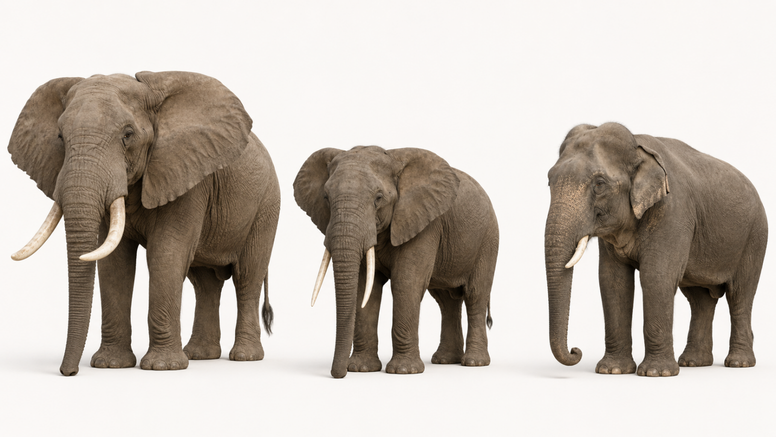
Classification in Marketing

- **Churn prediction:** Identify customers at risk of leaving.
- **Response modelling:** Predict whether a customer will respond to an offer.
- **Lead qualification:** Classify prospective customers by conversion likelihood.

Classification in Autonomous Driving

- **Object recognition:** Classify detected road users and obstacles.
- **Traffic-sign recognition:** Predict sign type from camera images.
- **Scene understanding:** Classify road, weather, and visibility conditions.
- **Driver monitoring:** Recognize distraction or fatigue states.

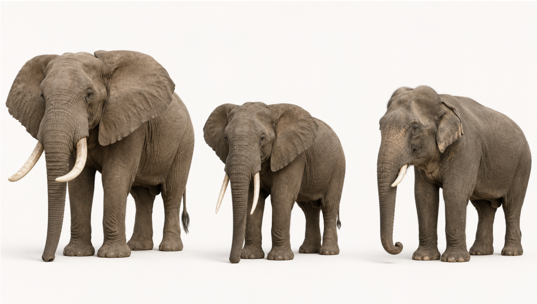
A Three-Class Classification Dataset



African savanna elephant　　African forest elephant　　Asian elephant

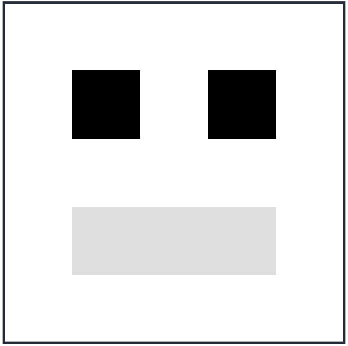
Feature Vectors (X)

Illustrative features (0–100): ear ratio, tusk angle, head-profile score

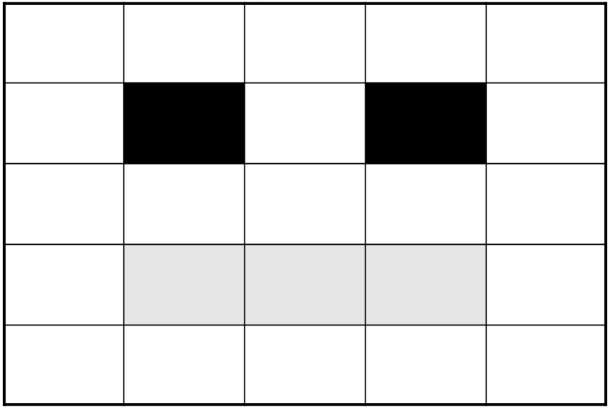


African Savanna Elephant	African Forest Elephant	Asian Elephant
$x_1 = (85, 42, 15)$	$x_2 = (68, 15, 20)$	$x_3 = (35, 55, 90)$

Data Representation



Data Representation

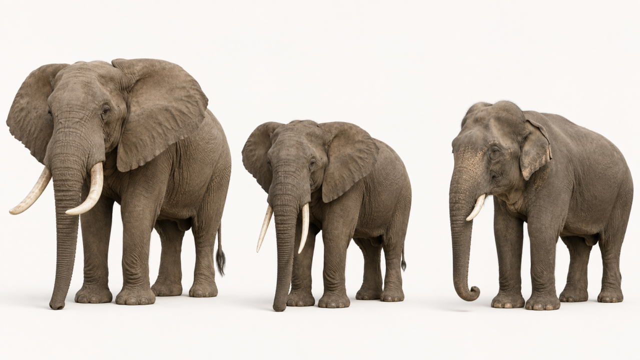


Data Representation

1	1	1	1	1
1	0	1	0	1
1	1	1	1	1
1	0.5	0.5	0.5	1
1	1	1	1	1

From Objects to Labels

Elephant images $\longrightarrow$ features X $\longrightarrow$ species Y



Classifying a New Elephant

$$x = (82, 40, 18)$$

$$h(82, 40, 18) = ?$$

Classifying a New Elephant

$$x = (65, 18, 22)$$

$$h(65, 18, 22) = ?$$

Classifying a New Elephant

$$x = (38, 52, 88)$$

$$h(38, 52, 88) = \, ?$$

Classifying a New Elephant

$$x = (88, 39, 12)$$

$$h(88, 39, 12) = ?$$

Training a Classifier

Training data (x_i, y_i) → Learning algorithm minimize loss → Fitted classifier $h_\theta(x)$ → Prediction $\hat{y} = h_\theta(x)$

Training chooses parameters

$$\hat{\theta} \in \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, h_\theta(x_i))$$

that minimize average loss on the observed examples.

Model Families

- **Linear classifiers:** Logistic regression, perceptron, linear SVM.
- **Generative classifiers:** LDA, QDA, and naive Bayes.
- **Local and kernel methods:** k-nearest neighbours, RBF models, kernel SVMs.
- **Trees and ensembles:** Decision trees, random forests, bagging, and boosting.
- **Neural networks:** Feedforward networks, CNNs, and modern deep architectures.

Choosing a Loss Criterion

Choose a criterion that matches the learning goal.

0–1 loss

Counts errors

Direct, but hard to optimize

Log loss

Rewards calibrated probabilities

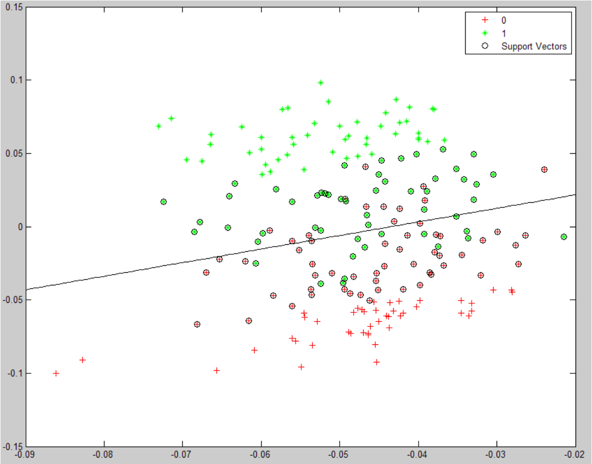
Smooth and widely used

Hinge loss

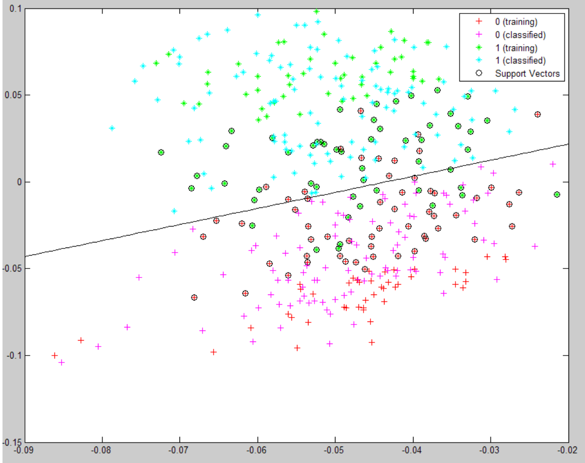
Encourages a large margin

Used by support vector machines

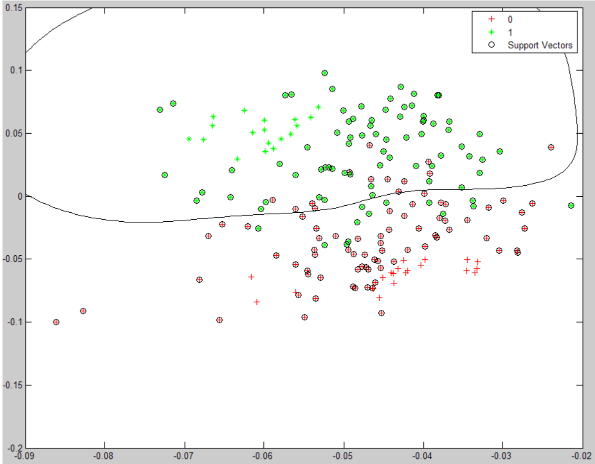
Linear Classification



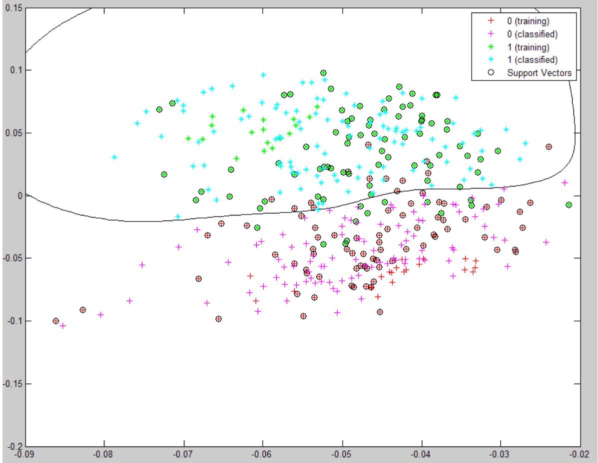
Multiclass Classification



Nonlinear Classification



Nonlinear Multiclass Classification



Handwritten-Digit Classification

The Matlab data file 2_3.mat contains 200 handwritten 2's and 3's images scanned from postal envelopes, like the ones shown below.

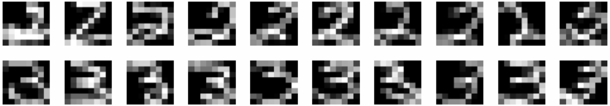
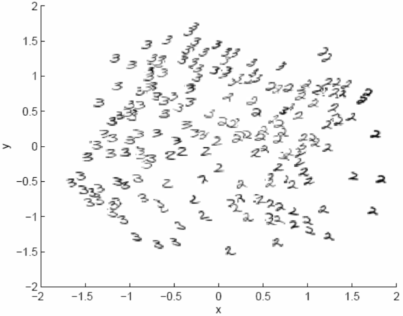


Figure 1:

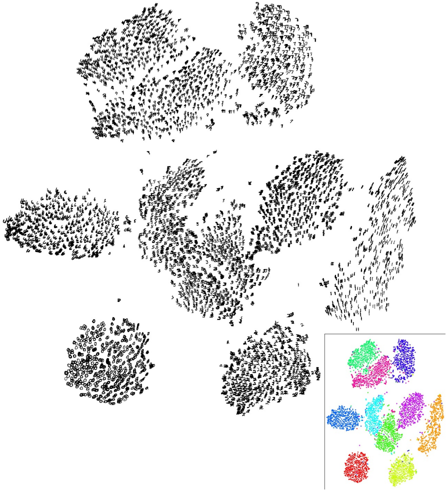
These images are stored as a 64×400 matrix. Each column of the matrix is an 8×8 greyscale image (the pixel intensities are between 0 and 1).

Representation Learning



A canonical dimensionality reduction problem from visual perception. The input consists of a sequence of 64-dimensional vectors, representing the brightness values of 8 pixel by 8 pixel images of digits 2 and 3. Applied to $n = 400$ *raw images. A two-dimensional projection is shown, with the original input images.*

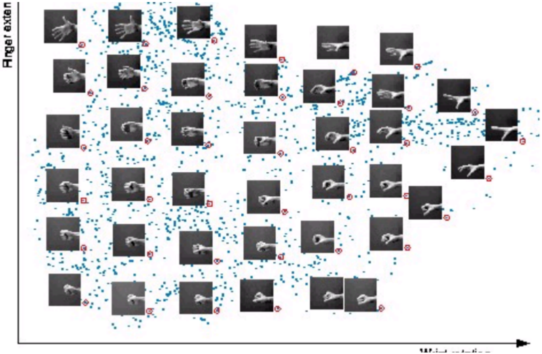
t-SNE Visualization



Interpreting t-SNE

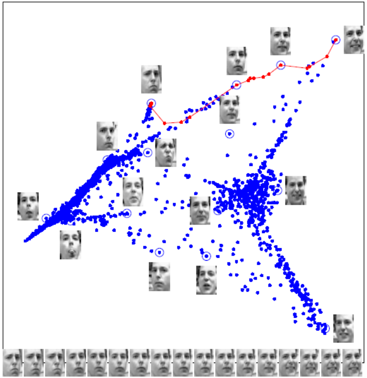
t-SNE can reveal local neighbourhood structure, but apparent clusters should be interpreted cautiously.

Isomap: Recovering a Nonlinear Manifold



Tenenbaum, de Silva, and Langford (2000)

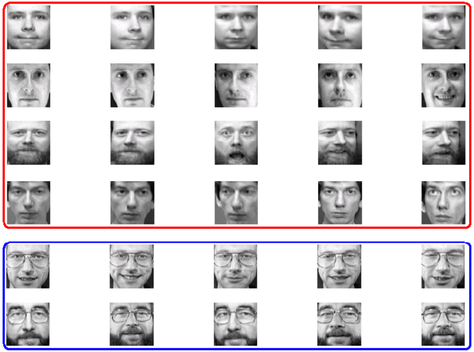
Isomap Embedding



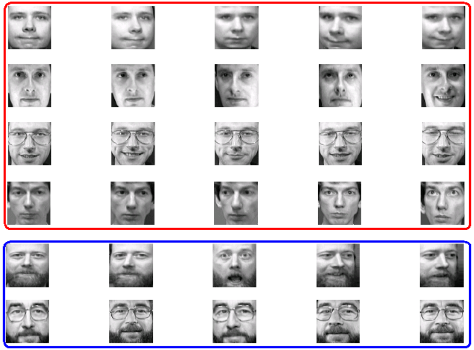
Images of faces mapped into the embedding space described by the first two coordinates of LLE, using $K = 12$ nearest neighbors. Representative faces are shown next to circled points at different points of the space. The bottom images correspond to points along the top-right path, illustrating one particular mode of variability in pose and expression. The data set had a total of $N = 1965$ grayscale images at 20×28 resolution ($D = 560$).

Representations and Attributes

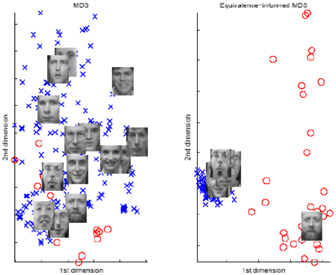
Separating Glasses vs. No Glasses



Separating Beard vs. No Beard

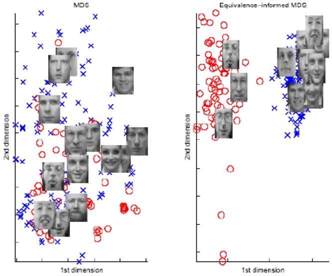


A Representation for Beard



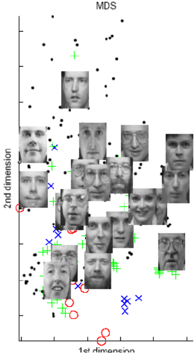
MDS and Equivalence-Informed MDS with bearded/unbearded distinction (all class-equivalent pairs)

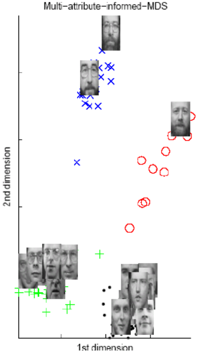
A Representation for Glasses



MDS and Equivalence-Informed MDS with glasses/no glasses distinction (all class-equivalent pairs)

Learning a Metric for Multiple Attributes





Textbooks and Course Materials

- **No mandatory textbook.** Lectures and posted materials define the course curriculum.
- **Recommended references:** Use the next slide as a guide for additional reading.

Recommended References

- Hastie, Tibshirani, and Friedman: *The Elements of Statistical Learning*.
- Bishop: *Pattern Recognition and Machine Learning*.
- Murphy: *Machine Learning: A Probabilistic Perspective*.

Assessment

Component	Structure	Weight
Assignments	Four individual assignments	50%
Paper presentation	Small-group research-paper presentation	10%
Group final project	Research or applied classification project	40%
Total		**100%**

Tentative Assessment Dates

Assessment	Release	Due
Assignment 1	Mon., Sept. 28	Mon., Oct. 19, 11:59 p.m.
Assignment 2	Mon., Oct. 19	Mon., Nov. 9, 11:59 p.m.
Assignment 3	Mon., Nov. 9	Mon., Nov. 23, 11:59 p.m.
Assignment 4	Mon., Nov. 23	Mon., Dec. 7, 11:59 p.m.
Paper presentation	Selection process TBA	Scheduled presentation
Group final project	–	Mon., Dec. 21, 11:59 p.m.

Paper Presentation (10%)

- **Work in a small group:** Every member must participate and answer questions.
- **Select an eligible research paper:** Venues and the selection process will be announced.
- **Explain the contribution:** Motivation, method, evidence, limitations, and connections to the course.
- **Schedule:** Presentation dates and the evaluation rubric will be announced.

Assignments (50%)

- **Four individual assignments:** Each assignment is worth 12.5%.
- **Expected work:** Mathematical derivations, proofs, programming, data analysis, and computational experiments.
- **Submission:** Follow the instructions and submit through Learn.
- **Understanding matters:** You may be asked to explain part of your work orally or in a short quiz.

Group Final Project (40%)

- **Goal:** Investigate a substantive classification problem, method, or research question.
- **Deliverable:** A final report, supported by reproducible analysis or experiments.
- **Originality:** The project must be new work for this course.
- **Due date:** Monday, December 21, 2026, at 11:59 p.m.

Assessment Practices

- **Dates are tentative:** Approved changes will be announced through official course platforms.
- **Reproducibility:** Submit readable code, document key choices, and set random seeds where relevant.
- **Authorship:** Be prepared to explain submitted work and disclose substantive generative-AI use.
- **Integrity:** Cite external ideas, text, code, data, and figures appropriately.

Communication

- **Use Piazza for course questions:** Public posts allow everyone to benefit from the discussion.
- **Use private posts when needed:** Personal matters or questions that reveal solution details should not be public.

Using Piazza Effectively

- **Search first:** Check whether the question has already been answered.
- **Use a descriptive subject:** Make the question easy for others to find.
- **Tag and keep it concise:** Help classmates and course staff respond efficiently.

Academic Integrity on Piazza

- **Do not reveal solutions:** Avoid posting assignment or project details that enable copying.
- **Use a private post:** Share specific code, derivations, or attempted solutions only with course staff.
- **When uncertain:** Ask before posting.

Course Platforms and Contacts

- **Course website:** https://aghodsib.github.io/courses/stat841/
- **Piazza:** Access through the widget on the main course webpage.
- **Learn:** Submission dropboxes and grades.
- **Instructor office hours:** Wednesdays, 1:00–2:00 p.m., M3 4009.
- **TA:** Nafisat Ibrahim, n4ibrahi@uwaterloo.ca; office hours TBA.

Prerequisites

- **Graduate students in Statistics, Computer Science, ECE, or SYDE:** No formal prerequisite.
- **Students from other programs:** Instructor permission is required.

Tentative Topics I

- **Foundations:** Loss, risk, decision rules, Bayes classifier, and error rates.
- **Gaussian and linear classifiers:** LDA, QDA, covariance modelling, and discriminant functions.
- **Linear and logistic models:** Likelihood, optimization, and regression views of classification.
- **Feature selection and representation:** Dimensionality reduction and feature extraction.
- **Evaluation and model selection:** Validation, metrics, calibration, complexity, and generalization.
- **Nearest neighbours and metric learning:** Local rules, learned distances, and embeddings.

Tentative Topics II

- **Support vector machines and kernels:** Margins, soft margins, duality, and nonlinear boundaries.
- **Naive Bayes and decision trees:** Conditional independence, splitting, pruning, and interpretation.
- **Bagging and boosting:** Random forests, variance reduction, and adaptive boosting.
- **Radial basis function networks:** Localized basis functions for classification.
- **Neural networks and deep learning:** Backpropagation, regularization, and deep classifiers.
- **Semi-supervised and emerging methods:** Unlabelled data, robustness, and selected topics.